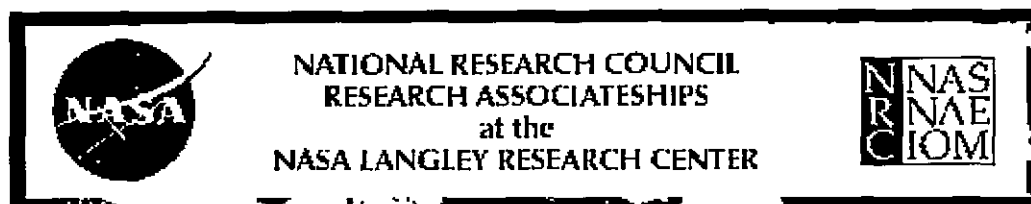


Langley Research Center

Resident Research Associateships

Postdoctoral and Senior Research Awards 1997



OPPORTUNITIES FOR RESEARCH

tenable at the

LANGLEY RESEARCH CENTER
Hampton, Virginia

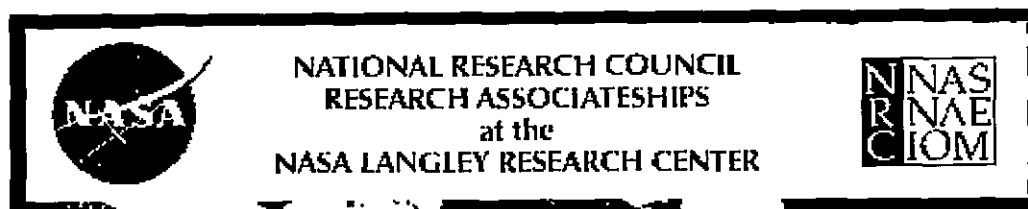
administered by the
NATIONAL RESEARCH COUNCIL
Washington, DC

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NAS

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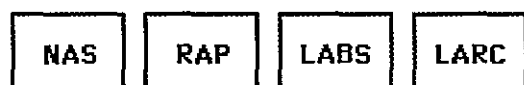


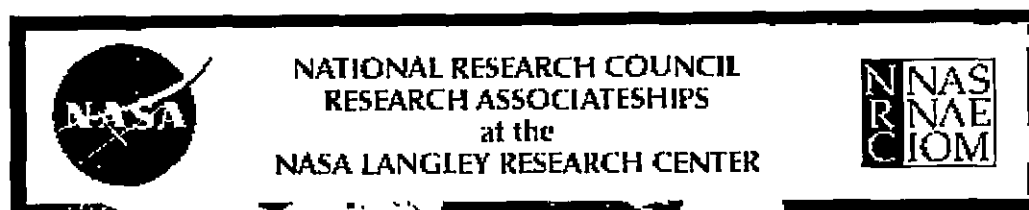
Foreword

The role of the Langley Research Center is to conduct innovative research necessary to advance aeronautical and space flight; to generate advanced concepts for the accomplishment of related national goals; and to provide research advice, technological support, and assistance to other NASA installations, other government agencies, and to industry.

The aeronautical research program is aimed at identifying and pursuing basic and applied opportunities for research that offer the greatest potential for increases in performance, efficiency, and capability. Research at Langley addresses both civil and military needs and spans the complete flight spectrum of subsonic, transonic, supersonic, and hypersonic speeds. The aeronautics program is accomplished with extensive research laboratories, full-scale aircraft, and an outstanding computational capability. Among the research laboratories are a variety of wind tunnels covering the entire Mach-number and Reynolds-number range. The most recent addition to Langley is the National Transonic Facility, which is a new, cryogenic, high-pressure wind tunnel providing a unique opportunity for conducting high-Reynolds-number research at subsonic and transonic speeds. Major research disciplines include aerodynamics; operations and airworthiness; acoustics and noise reduction; structures and materials; flutter, aeroelasticity, dynamic loads, and structural response; fatigue and fracture; electronic and mechanical instrumentation; computer technology; flight dynamics and control; and communications technology.

In keeping with the nation's space objectives, Langley maintains a broad interdisciplinary space-research program to provide the advanced scientific and technical knowledge necessary for manned and unmanned exploration and utilization of space. This program provides advanced research to develop new vehicle concepts, new design principles, and new operational techniques leading to more effective space and aerospace systems. The program includes the study of advanced aerospace transportation systems, large space-antenna systems, and a permanent manned space station. Langley also carries on a vigorous program to apply existing and emerging space technologies and capabilities to the needs and problems of society. This effort focuses on theoretical and experimental atmospheric sciences including stratospheric processes, global geochemical cycles, the Earth radiation budget, and the development and demonstration of techniques for the remote monitoring of environmental quantities.





Introduction

History and Objectives

The National Research Council conducts the Research Associateship Programs in cooperation with sponsoring federal laboratories and research organizations approved for participation.

The Research Council, through its Associateship Programs office, conducts a national competition to recommend and make awards to outstanding scientists and engineers at recent postdoctoral and experienced senior levels for tenure as guest researchers at participating laboratories. These Programs have been conducted on behalf of a number of federal agencies since 1954.

The objectives of the Programs are (1) to provide postdoctoral scientists and engineers of unusual promise and ability opportunities for research on problems, largely of their own choice, that are compatible with the interests of the sponsoring laboratories and (2) to contribute thereby to the overall efforts of the federal laboratories.

For recent doctoral graduates, the Programs provide an opportunity for concentrated research in association with selected members of the permanent professional laboratory staff, often as a climax to formal career preparation.

For established scientists and engineers, the Programs afford an opportunity for research without the interruptions and distracting assignments of permanent career positions.

Participating laboratories receive a stimulus to their programs by the presence of bright, highly motivated, recent doctoral graduates and by senior investigators with established records of research productivity. New ideas, techniques, and approaches to problems contribute to the overall research climate of the laboratories. Indirectly, Associateships also make available to the broader scientific and engineering communities the excellent and often unique research facilities that exist in federal laboratories.

For the 1997 program year, an anticipated 1,300 applications will be received for the nearly 350 new awards to be made in the Associateship Programs.

Associates on Tenure

In the programs sponsored by the Centers of the National Aeronautics and Space Administration (NASA), a Research Associate is a guest researcher, not an employee of the Research Council or of the Center, and there is no obligation to either organization during or after the completion of tenure.

Associateships are analogous to fellowships or similar temporary programs at the postdoctoral level in universities and other organizations. They are not intended to be, or to compete with, permanent professional career positions. No commitment on the part of an Associate, the sponsoring Center, or the Research Council with regard to later employment is implied or should be inferred by the offer or acceptance of an award.

Associates must devote their full-time effort to the research program proposed in their applications and must be in residence at the sponsoring Center during the entire period of the Associateship. No period of tenure may be spent in residence at another laboratory or institution. Associates have the status of visiting scientists or engineers but are subject to the general regulations of the Center. No additional monetary aid or other remuneration may be accepted from another appointment, fellowship, or similar grant, except for sabbatical leave, during the period of the Associateship.

Postdoctoral or Senior Research Associate Status and Length of Tenure

Postdoctoral Research Associateships are awarded to persons who have held the doctorate less than five years at the time of application and are made initially for one year.

Senior Research Associateships are awarded only to applicants who have held the doctorate five years or more at the time of application. Senior Research Associateship applicants should have research experience that has resulted in significant contributions and recognition as established investigators in their specialized fields. Although awards to Senior Research Associates are usually for one year, awards for periods of three months or longer will be considered.

Under certain conditions, extensions may be granted to allow Associates to bring their research to a reasonable stage of completion. However, extensions are not automatically granted, and applicants are advised to plan their research programs to conform to the length of tenure stated above.

Consideration

Qualified applicants receive consideration without regard to race, creed, color, age, sex, or national origin.

Citizenship

All opportunities for research at NASA Centers are open to citizens of the United States and to legal Permanent Residents.

Non-US citizens may submit proposals in basic science areas only and must have full command of the English language.

However, at some Centers, certain opportunities for research are open only to US citizens and Permanent Residents, and they are so designated by a statement at the beginning of the opportunity text.

Although opportunities are limited to unclassified research, all guest investigators, including Research Associates, are subject to NASA regulations governing visiting scientists.

NASA cannot sponsor applicants for its Research Associateship programs from citizens of countries whose policies have been determined to be inimical to US interests.

Prospective non-US applicants who are uncertain of their eligibility are advised to contact the Associateship

Programs office before submitting a formal application.

Visa Requirements

Non-US nationals who are offered awards must have valid visas throughout tenure. Only Exchange Visitor and Immigrant Visas are acceptable to the Research Council. If an awardee chooses to apply for an Exchange Visitor Visa, sponsorship must be by the National Research Council. If he or she chooses to apply for an Immigrant Visa, the Research Council is not involved in the procedure.

Education and Experience

Awardees must hold the PhD, ScD, or other earned research doctoral degree recognized in US academic circles as equivalent to the PhD or must present acceptable evidence of having completed all the formal academic requirements for one of these degrees before tenure may begin. Applicants must have demonstrated superior ability for creative research.

An applicant's training and research experience may be in any appropriate discipline or combination of disciplines required for the proposed research.

Opportunities for Research at the Langley Research Center (Langley)

This booklet contains abstracts, or opportunities for research, that describe areas of research in which Associateships may be awarded at the Langley Research Center (Langley).

Langley provides the funds for this program and furnishes all necessary support services, facilities, and equipment for the approved research program of each Associate.

While every effort has been made by Langley to provide opportunities of ample scope and relevance, the publication of any opportunity in this booklet does not guarantee that it will be available at the time awards are offered. Changes and/or deletions may occur following publication because of reorganization, temporary lack of equipment, laboratory renovation, staffing already sufficient to meet research goals, or a lack of funding.

Research Adviser and Center Program Representative

Shown with each opportunity for research are the names of one or more Research Advisers who conduct or direct the work described in the opportunity.

An Adviser is a scientist or engineer at Langley with whom a Postdoctoral Research Associate works most closely. An Adviser acts as a surrogate of the Research Council in monitoring an Associate, and all matters relating to an Associate's research program fall under his or her purview.

For a Senior Research Associate, an Adviser functions in a more collegial relationship and assists as needed in securing technical support and resources.

The Center Program Representative is a professional staff member at Langley who is responsible for managing its Research Associateship program and for assisting an Associate with all administrative aspects of tenure:

Roger Hathaway

University Affairs Officer
Office of Education
NASA LANGLEY RESEARCH CENTER
Mail Stop 400, Building 1216T
Hampton, Virginia 23681-0001
Telephone: (804) 864-4000
E-Mail: r.a.hathawy@larc.nasa.gov

Research Proposal

Each applicant must submit a research proposal that relates to a specific opportunity for research at Langley. A proposal must be the original work of an applicant and be approved by an Adviser listed with the opportunity.

Before writing a proposal, however, an applicant is advised to communicate directly with the Adviser, who can provide more specific information on current research and available technical facilities and offer scientific support of proposal development.

Laboratory/Center Review

Each applicant's proposal must be approved by one of the Advisers listed in this booklet and endorsed by the Program Committee of Langley to be eligible for an award.

The endorsement affirms that the proposal is compatible with Langley's interests and that adequate programmatic support will be available if an award is offered.

Langley's action on the proposal, together with a copy of the Adviser's comments, will be provided directly to the applicant by the Program Representative.

No applicant will be eligible for further consideration until the Associateship Programs office has been advised by Langley that his or her proposal has been approved by an Adviser and endorsed by the Program Committee. Otherwise, the Associateship Programs office will assume that the proposal is not of sufficient current interest to Langley or that support facilities cannot be made available.

Since the final review of applications is conducted by special panels appointed by the Research Council, all applicants should note that endorsement by an Adviser or Center, while essential to the application process, does not imply or guarantee an award by the Research Council.

The Panel Review

The Associateship Programs office receives all application materials and supporting documents and conducts the competitive evaluations of applications.

Evaluations are conducted in February, June, and October by special panels convened for this purpose. Panelists are chosen to review applications on the basis of their stature and experience in the fields of science and engineering, and their evaluations become the basis from which awards are made on behalf of the Langley.

Applicants are recommended for awards only after this open, national competition in which the panels rank candidates on the basis of quality alone.

Final ranking in order of quality and the recommendation of applicants for awards are the exclusive prerogatives of the panels, and only notification by the Associateship Programs office of an applicant's status in the competition is authoritative.

Stipend

An Associate receives a stipend from the Research Council while carrying out his or her proposed research. The current annual stipend for a Postdoctoral Research Associate is \$34,500. An appropriately higher stipend will be offered to Senior Research Associates.

Stipends are subject to adjustments from time to time in accordance with general national guidelines pertaining to scientists and engineers.

Following current salary practices in certain professional fields that are experiencing a short supply of new doctoral graduates, a stipend supplement may be allotted by the Research Council to awardees in those disciplines.

For the 1997 program year, supplements of up to \$6,500 may be added to the basic stipend of Postdoctoral Research Associates holding recognized doctoral degrees in engineering and computer science.

The Research Council is required by the US Tax Code to withhold an amount from the stipends of non-resident aliens who hold Exchange Visitor (J-1) Visas. Exchange Visitors are advised that approximately 30% per month will be withheld from stipends and reported to the US Internal Revenue Service annually.

Applicants are cautioned against entering into any agreement or understanding with individual Advisers or other Center personnel for additional funding or other remuneration for work as an Associate.

Stipends for Associates are limited to the amounts and by the conditions set forth above, and any other arrangement, formal or informal, between an applicant and Center personnel for additional monies or other considerations is strictly prohibited by the Research Council.

Initiation of Tenure

Sufficient time must be allowed between the offer of an award and the beginning of tenure to enable the Associateship Programs office and Langley to complete all necessary administrative procedures.

The date on which tenure may begin is negotiated on an individual basis, normally within six months of the award. The starting date may be delayed by mutual agreement of Langley, the Associate, and the Associateship Programs office but cannot be later than 12 months from the date on which the award was originally offered.

If this condition cannot be met, a new application, including a newly approved research proposal, must be submitted to the Associateship Programs office and will be judged without prejudice in the next competition.

Prior Affiliation with the Center

A primary objective of the Associateship Programs is to provide a mechanism for new ideas and sources of stimulation to be brought to the sponsoring Center. Thus persons with recent prior affiliation with a specific

Center may not be eligible to apply for an Associateship there.

Prior affiliation includes direct employment relationships either with the Center or with a contractor whose work is performed there. A long-term consulting relationship usually makes an applicant ineligible.

Research contracts with universities that provide support for graduate students or faculty who perform research on campus are not ordinarily considered to be disqualifying.

Reapplication

Persons who have previously held an Associateship may apply for another award only if a period of at least two years will have elapsed between termination of the first award and the proposed tenure of a second.

Persons who have previously applied for an Associateship, but who were not recommended for an award by the panels, may reapply after one year.

Candidates who were recommended for an award by the panels, but who were not offered an award because of funding or other limitations, may reapply at any time without a mandatory waiting period.

Taxes and Insurance

As a guest investigator, an Associate is self-employed. All arrangements for payment of income taxes are the responsibility of the individual Associate, who is advised to become familiar with the relevant sections of the current tax codes.

The Research Council is required by the US Tax Code to withhold an amount from the stipends of non-resident aliens who hold Exchange Visitor (J-1) Visas. Exchange Visitors are advised that approximately 30% per month will be withheld from stipends and reported to the US Internal Revenue Service annually.

Job-related injury or death is covered by insurance (workmen's- compensation type). A group health-insurance program is required for Associates and is optional for dependents.

Relocation and Travel

A suitable relocation reimbursement is determined for each awardee.

Funds are also available for limited professional travel during tenure, provided such travel is approved in advance by the Associate's Adviser, the Langley Program Representative, and the Associateship Programs office. Details will be provided at the time of the award.

Publication

Since an Associate's later scientific and technical career will be judged by others, publication in the accepted open technical literature is highly encouraged.

Publications should include a statement indicating that the research was conducted while the author held a National Research Council Research Associateship.

Application Procedure

Detailed information on procedures and all necessary application materials and supporting documents are available on request from the

Associateship Programs
TJ 2114
National Research Council
2101 Constitution Avenue NW
Washington, DC 20418
E-Mail: rap@nas.edu
WWW (Internet): <http://rap.nas.edu>

All deadlines for receipt of application materials are strictly observed by Langley and the Associateship Programs office. No allowances or exceptions are made for late submissions. Application materials from previous competitions may not be used.

Panel Review Schedule

Although applications for Langley Research Associateships are accepted throughout the year, they are evaluated by the panels only during February, June, and October.

February Review

To be eligible for review in February, completed application materials must be postmarked no later than January 15, 1997, and received by the Associateship Programs office no later than January 25, 1997. Supporting documents must be received by February 15, 1997.

June Review

To be eligible for review in June, completed application materials must be postmarked no later than April 15, 1997, and received by the Associateship Programs office no later than April 25, 1997. Supporting documents must be received by June 1, 1997.

October Review

To be eligible for review in October, completed application materials must be postmarked no later than August 15, 1997, and received by the Associateship Programs office no later than August 25, 1997. Supporting documents must be received by October 1, 1997.

Application Materials

Submit the following to the Associateship Programs office:

Signed Application
Questionnaire Sheet
Research Proposal (1 Copy)
Previous and Current Research

Supporting Documents

Have the following sent directly to the Associateship Programs office:

For Postdoctoral Research Associateship Applicants

Official transcripts of all graduate and undergraduate credits.

Four Reference Reports from the respondents listed on the Application. Only official Reference Reports may be used.

For Senior Research Associateship Applicants

Transcripts are not required of Senior Research Associateship applicants.

Letters of reference are accepted in lieu of Reference Reports. Senior Research Associateship applicants should endeavor to include some references from individuals who are not co-employees.

Laboratory/Center Documents

Submit the following directly to the Center Program Representative:

Research Proposal (1 Copy)

Laboratory/Center Review (1 Set)

The Program Representative will forward the documents to the Adviser and the Program Committee for their reviews.

Notification of Awards

Awards are made only by the National Research Council. The endorsement of an application and research proposal by Langley, while essential to the application and review processes, does not constitute an agreement or obligation to confer an award.

A review board, drawn from members of the Research Council panels, determines a cutoff score. Applicants who score below this score cannot be considered further for an award and are so notified within two to four weeks.

Applicants who score above the cutoff score are recommended for awards by the board. These applicants are notified of the board's action as early as possible following the review and are offered awards or alternate status to the extent of available facilities and funding by Langley.

Acceptances and declinations must be made directly to the Associateship Programs office of the National Research Council.

NASA Centers Participating in the Research Associateship Programs

Ames Research Center, Moffett Field, California

Dryden Flight Research Center, Edwards, California

Goddard Space Flight Center, Greenbelt, Maryland, also with sites at:

Goddard Institute for Space Studies, New York, New York
Wallops Flight Facility, Wallops Island, Virginia

Jet Propulsion Laboratory, Pasadena, California

Lyndon B Johnson Space Center, Houston, Texas

John F Kennedy Space Center, Kennedy Space Center, Florida

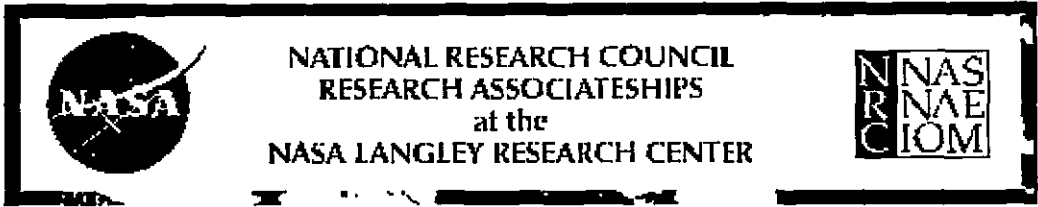
Langley Research Center, Hampton, Virginia

Lewis Research Center, Cleveland, Ohio

George C Marshall Space Flight Center, Marshall Space Flight Center, Alabama

John C Stennis Space Center, Stennis Space Center, Mississippi





Remote Sensing

- ☐ Inversion and Analysis of Remote-Sensing Data
- ☐ LIDAR Atmospheric Measurements of Water Vapor
- ☐ Satellite, Aircraft, and Ground Observed Cloud and Radiative Flux Analyses



NRC Research Associateship Programs

Langley Research Center

Inversion and Analysis of Remote-Sensing Data

RO#: 44.50.24.01

CITY: Hampton
STATE: VA 23681-0001

ADVISER: Chu, William P (804)864-2675

KEYWORD: Aerosols, Atmospheric chemistry, Atmospheric moisture, Atmospheric ozone, Atmospheric remote sensing, Lidar, Satellite atmospheric observations, Stratosphere, Troposphere

Basic and applied research is being conducted in the development of advanced techniques for retrieving and analyzing atmospheric data from satellite, airborne, and ground-based remote-sensing instruments. Techniques are being developed for processing satellite atmospheric limb scattering and laser-radar measurements, data from existing satellite solar occultation experiments (SAM II, SAGE, SAGE II, and SAGE III), and various LIDAR measurements. Our goal is to retrieve stratospheric and tropospheric constituent concentration profiles and study their influence on atmospheric chemistry and dynamics. Current investigations include the retrieval of tropospheric and stratospheric water vapor, ozone, nitrogen dioxide, and aerosol information from solar extinction and LIDAR data; an assessment of the errors associated with the determination of global ozone trends using solar occultation measurements from satellites; the measurement of atmospheric constituent absorption cross sections at various visible wavelengths for use in converting constituent extinction to concentration information; and the determination of various aerosol optical properties from multiwavelength data. Numerical studies are supported with programming assistance and by a large range of computer facilities (CONVEX, CRAY-2S, SUN, and DEC workstations).

PANEL: Physics

You may use the following form to add this opportunity to your database or check all the opportunities you have saved in your database. To do that, you need to provide the correct username and password first.

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NRC Research Associateship Programs

Langley Research Center

LIDAR Atmospheric Measurements of Water Vapor

RO#: 44.50.24.02

CITY: Hampton
STATE: VA 23681-0001

ADVISER: Browell, Edward Vern (804)864-1273

KEYWORD: Atmospheric moisture, Atmospheric remote sensing, Lidar, Satellite atmospheric observations, Stratosphere, Troposphere

LIDAR techniques that have important applications in areas of meteorology and atmospheric sciences are being developed for use from airborne and eventual spaceborne platforms. The first steps have been taken in this development by building and flight testing differential absorption LIDAR (DIAL) systems operating in the 720- and 815-nm wavelength regions. These systems use either Alexandrite lasers or Nd:YAG-pumped Ti:Sapphire lasers to produce tunable, narrow-line laser operation for DIAL water vapor measurements. The capability of these airborne DIAL systems will be extended to the 940-nm region, where water vapor absorption lines have line strengths a factor of 20 greater than those in the 720- or 815-nm regions. The increased water vapor absorption will permit investigation of lower water vapor concentrations in the upper troposphere and lower stratosphere. Different laser techniques for generating 940-nm laser wavelengths will be explored such as Raman shifting of the Alexandrite laser output, the use of Cr:LiSAF lasers, and OPO wavelength shifting of various lasers. Research will include laboratory investigations of laser techniques for generating laser wavelengths in regions of strong water lines. Ground-based DIAL measurements will be performed, and airborne evaluation of potential DIAL laser candidates for water vapor measurements in the 940-nm region will also be conducted. An assessment of these techniques for future spaceborne water vapor experiments will be performed.

PANEL: Earth Sciences

You may use the following form to add this opportunity to your database or check all the opportunities you have saved in your database. To do that, you need to provide the correct username and password first.

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Password

Pick an option



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RAP

LABS

LARC

NRC Research Associateship Programs

Langley Research Center

Satellite, Aircraft, and Ground Observed Cloud and Radiative Flux Analyses

RO#: 44.50.24.03

CITY: Hampton

STATE: VA 23681-0001

ADVISER: Wielicki, Bruce Anthony (804)864-5683

KEYWORD: Atmospheric radiation, Atmospheric remote sensing, Clouds, Lidar, Satellite atmospheric observations, Microwave remote sensing

Basic and applied research is being conducted in remote sensing of the Earth's cloudiness and radiative fluxes (top of atmosphere, surface, within atmosphere). Clouds are the primary modulators of the radiative energy balance of the Earth's surface and atmosphere on both local and global scales. Data sets from several satellite instruments, including Geostationary Operational Environmental Satellite, Landsat, AVHRR, HIRS, SSM/I, Earth Radiation Budget experiment, and Nimbus 7 will be analyzed and compared. In addition, simultaneous aircraft and/or ground-based laser-radar measurements will be used to examine cloud cover, cloud base and top height, optical depth, reflectance, cloud particle size, liquid water path, and emissivity. Comparisons will be made with theoretical predictions of cloud generation/dissipation and with models of cloud radiative properties such as cloud albedo, bidirectional reflectance, and emissivity. Simulation studies will examine the sampling requirements and cloud measurement capabilities of current and future satellite measurement systems. Use of passive and active microwave observations are of special interest in addressing the problems associated with multilayered cloud systems.

PANEL: Earth Sciences

You may use the following form to add this opportunity to your database or check all the opportunities you have saved in your database. To do that, you need to provide the correct username and password first.

Username

Password

Pick an option

